Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2026
Team ID	LTVIP2026TMID00420
Project Name	EV Market Analysis Using Tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	System shall access Vahan Parivahan dataset (17M rows) from Kaggle, with EV sales data by state, year, maker	High
2	Data Processing	System shall clean data - remove nulls, filter only EV category (2W, 3W, 4W), convert date formats	High
3	Business Analysis	System shall create KPIs - Total EV sales, Market share %, YoY growth, Top states & makers analysis	High
4	Business Insights & Analytical Outputs	System shall generate dashboards - State map, Year-wise line chart, Maker bar chart, Category pie chart	High
5	User Interaction & Analysis	Users shall filter dashboard by Year, State, Vehicle Category, Maker and drill down for detailed analysis	Medium
6	Data Update & Reporting	System shall allow monthly data update and generate Tableau Story with 5 slides of key insights	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Dashboard should be simple, user-friendly, easy for non-technical users to understand
NFR-2	**Security**	Vahan sensitive registration data should be protected, only authorized access
NFR-3	**Reliability**	Dashboard should give accurate and consistent results without errors
NFR-4	**Performance**	Dashboard should load within 5 seconds even with 17M rows data
NFR-5	**Availability**	Tableau Public dashboard should be available 24x7 with 99% uptime
NFR-6	**Scalability**	System should handle future data growth (up to 50M rows) without crashing